Addressing the $p \gg n$ Problem: A Study of Model Efficacy using Maize Flowering Time Prediction

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***Abstract*—This study aims to evaluate and compare $p \gg n$-compatible statistical models for predicting Maize Flowering Time within a high-dimensional setting. The objective is to identify the modeling method that delivers the optimal trade-off among predictive accuracy, model robustness against overfitting, and practical computational efficiency in handling large-scale data. This analysis will inform the selection of robust and efficient methods for a high-dimensional problem.**

I. Introduction

This report evaluates the efficacy of specialized statistical methodologies against a high-dimensionality genomic dataset for the prediction of Maize Flowering Time (dtoa). Traditional statistical approaches are inherently unsuitable for this problem setting, defined by the $p \gg n$ regime where the number of predictors (p) vastly outnumbers the observations (n).

A key consequence of high dimensionality is the instability of variance, which compromises the reliability and confidence of model performance due to an elevated risk of overfitting. Therefore, the chosen models must incorporate regularization and feature selection to stabilize estimates and identify true signals.

The core objective of this analysis is to determine which modeling method provides the optimal balance of predictive accuracy and robustness given compute time when faced with this complex, high-dimensional structure.

A. *Dataset Structure*

The dataset [1] consists of $4,981$ observations (n) and $7,390$ variables (p), including some amount of missing data. The dependent variable is Time to Male Flowering (`dtoa`). The independent variables are predominantly $7,389$ Single Nucleotide Polymorphism (SNP) markers ($m_1 - m_{7389}$), supplemented by 25 (1-25) categorical variables representing the genetic crosses or families (`pop`).

II. Methods

A. *Data Preparation*

*1) **Preprocessing**:* Initial data cleaning involved the *removal of incomplete observations*; specifically, any row containing one or more missing values was excluded from the effective dataset. This procedure reduced the total number of records from $4,981$ to $4,494$.

The resulting data loss, approximately 10% of the original sample, was deemed acceptable given the scope and objectives of this analysis. Following this cleaning step, the $4,494$ retained observations were *randomly partitioned* into three distinct subsets for model development and evaluation.

The subsets were allocated as follows: 75% ($\approx 3,371$ records) for the *training set*, 15% (≈ 674 records) for the *validation set*, and the remaining 10% (≈ 449 records) for the final, unbiased *test set*. This division ensures a rigorous assessment of model performance and generalization capabilities.

*2) **Standardization**:* To prepare the data for models designed to handle high-dimensionality, all features were subjected to *standardization*, often referred to as Z-score normalization. This critical preprocessing step transformed each feature such that its distribution had a mean of zero ($\mu = 0$) and a standard deviation of one ($\sigma = 1$). Mathematically, this transformation is defined as $z = (x - \mu)/\sigma$.

This procedure ensures that all features are centered around zero and scaled to unit variance, a step essential for promoting *equitable contribution* from every feature during the model's learning process.

B. *High-Dimensionality Regression Models Used*

1) **Stepwise Regression**: In particular the forward algorithm, this technique builds a model by iteratively adding or removing predictors based on a statistical metric, retaining only the most significant features. It works in $p \gg n$ scenarios by drastically reducing the effective number of predictors, thus avoiding the computational and statistical pitfalls of a full model.
2) **Lasso Regression**: Lasso adds an ℓ_1 penalty (absolute value of coefficients) to the model, which forces the coefficients of less important features to become exactly zero. This automatically performs crucial feature selection, making it excellent for $p \gg n$ by providing a sparse, interpretable model.

3) **Ridge Regression**: Ridge adds an ℓ_2 penalty (squared value of coefficients) to the model, which shrinks all coefficients towards zero but never sets them to exactly zero. This effectively stabilizes the model and mitigates the high variance and multicollinearity issues common in $p \gg n$ datasets.
4) **Elastic Net Regression**: Elastic Net combines both the ℓ_1 (Lasso) and ℓ_2 (Ridge) penalties in its objective function. It leverages the feature selection ability of Lasso while maintaining the stability of Ridge, making it robust for complex high-dimensional data with highly correlated predictors.
5) **Principal Component Regression (PCR)**: PCR first uses Principal Component Analysis (PCA) to transform the original p features into a smaller set of uncorrelated latent components (composite variables). It then runs an ordinary linear regression on these few components, completely solving the $p \gg n$ problem by moving to a much lower-dimensional space.
6) **Partial Least Squares Regression (PLS)**: PLS constructs new latent variables that not only explain the variance in the predictors but also maximize their correlation with the response variable ($\mathbf{Y}$) or `dtoa` in this case. This method is effective in $p \gg n$ because its dimension reduction is supervised and specifically optimized for predictive power.

C. *Tuning and Selection*

All models, specifically **Lasso**, **Ridge**, **Elastic Net**, **PCR**, and **PLS**, underwent hyperparameter optimization using *k-fold cross-validation* applied exclusively to the *training set*.

For PCR and PLS, the optimal number of components (k) was searched across the grid $\{5, 10, 50, 100, 200, 300, 400, 500\}$ using $k = 5$ folds. In contrast, the **Stepwise Regression** model utilized the Bayesian Information Criterion (BIC) as its selection metric due to BIC's stringent penalty on model complexity.

The optimal hyperparameters for the **Lasso**, **Ridge**, and **Elastic Net** models can be observed in Table I.

TABLE I
HYPERPARAMETER GRID FOR REGULARIZED REGRESSION MODELS

Grid Index	Lasso (α)	Ridge (α)	Elastic Net (α, l_ratio)
1	0.001	0.0001	0.001, $(0.1, 0.5, 1.0)$
2	0.0031622777	0.001	.01, $(0.1, 0.5, 1.0)$
3	0.01	0.01	0.1, $(0.1, 0.5, 1.0)$
4	0.0316227766	0.1	1.0, $(0.1, 0.5, 1.0)$
5	0.1	1.0	10.0, $(0.1, 0.5, 1.0)$
6	0.316227766	10.0	n/a
7	1.0	100.0	n/a
8	3.1622776602	1000.0	n/a
9	10.0	10000.0	n/a

Note: α is the regularization strength; l_ratio (L1 ratio) is the mixing parameter, where $0 < l\_ratio \leq 1$.

III. RESULTS

Analysis of the tuned models yielded distinct optimal configurations, as detailed in Table II.

TABLE II
PERFORMANCE AND COMPUTE TIME

Method	RMSE (CV)	RMSE (Test)	# Feats / Comps	Time (s)
Ridge	3.526	3.533	7390	136.914
Lasso	3.358	3.387	82	1214.324
Elastic Net	3.347	3.389	309	2026.851
Stepwise (BIC)	3.372	3.404	17	431.755
PCR (k comps)	3.542	3.593	50	141.986
PLS (k comps)	3.645	3.670	10	2927.664

For **Stepwise Regression**, the final model included 17 features, selected in forward order: {`pop`, `m439`, `m6493`, `m2419`, `m5808`, `m1652`, `m772`, `m290`, `m2548`, `m4940`, `m2969`, `m2865`, `m4772`, `m2162`, `m2276`, `m2358`, `m6665`}.

The regularized models achieved maximum performance with the following hyperparameters: **Ridge** utilized a strong penalty of $\alpha = 10000.0$; **Lasso** achieved optimal results at $\alpha = 0.1$, resulting in a sparse model with 82 non-zero coefficients; and **Elastic Net** also optimized at $\alpha = 0.1$ with an ℓ_1 ratio of 0.5 (leading to 309 non-zero coefficients).

Among the dimension reduction techniques, **PLS** performed best with 10 components, while **PCR** achieved its optimal fit using 50 components.

Across all models evaluated, the observed **Root Mean Square Error (RMSE)** from the cross-validation (CV) set was consistently lower than that of the independent test set, confirming expected model behavior and indicating a degree of generalization error common in predictive modeling.

While the time component reported in Table II is machine-dependent and thus serves primarily as a reference, the training times for the more complex models were substantial, often extending into the 40-minute range given the computational resources employed.

IV. DISCUSSION

A. *Models Performance*

A comparison of predictive metrics, visually represented in Figure 1, confirms that the **Lasso** and **Elastic Net** models achieved the superior performance, exhibiting the lowest prediction error, as indicated by a RMSE value of approximately 3.40, and the highest predictive power, with Q^2 values near 0.145.

Conversely, the **PLS** model performed the worst, recording the highest RMSE ($\approx$ 3.68) and the lowest predictive power ($Q^2 < 0.00$).

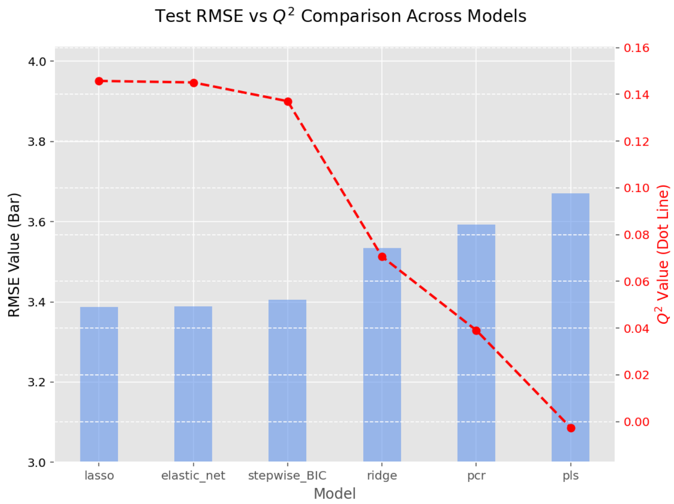

Fig. 1. Comparing Q^2 against RMSE

While the range of RMSE values across all models was narrow (3.40 to 3.68), the Q^2 metric clearly distinguished the superior predictive ability of the sparse regularization techniques (Lasso and Elastic Net).

As illustrated in Figure 2, revealed that although the RMSE and **Mean Absolute Error (MAE)** values were numerically close across all six models (RMSE clustered between 3.4 and 3.7; MAE between 2.6 and 2.9), their ratio provided a key distinction.

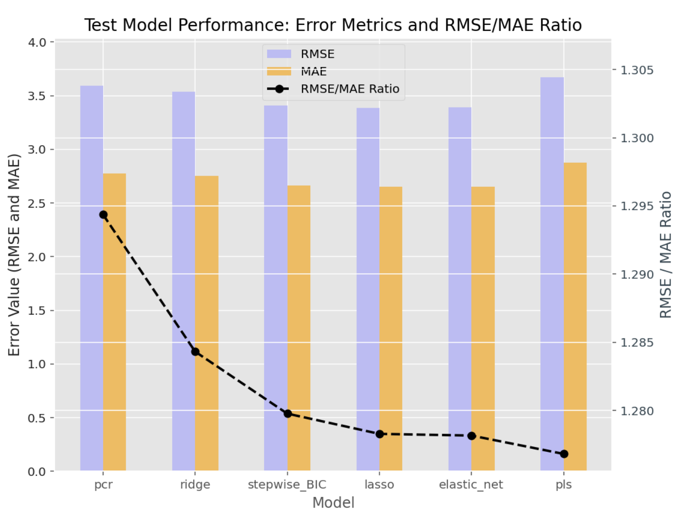

Fig. 2. Comparison of Regression Model Error and Ratio

RMSE/MAE ratio demonstrated a clear decreasing trend, ranging from ≈ 1.295 for the **PCR** model down to ≈ 1.275 for the **PLS** model.

This lower ratio for **PLS** suggests that, relative to the other models, the **PLS** approach minimized the influence of large outlier errors, indicating a more robust and uniform error distribution.

B. Accuracy-Time-Interpretability Tradeoff

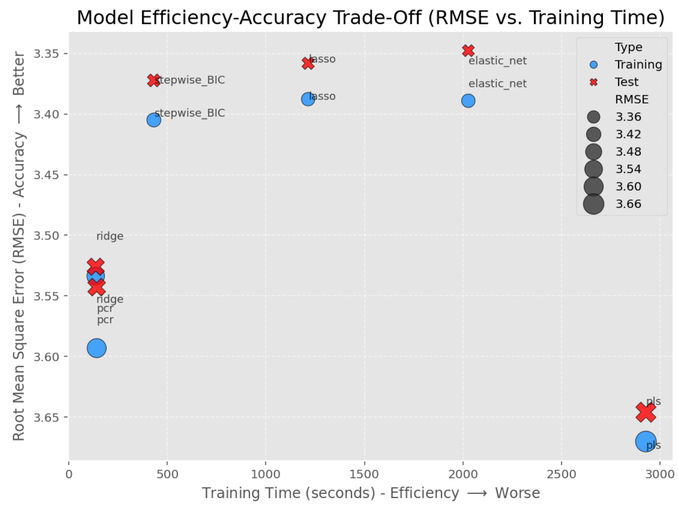

Fig. 3. Model Efficiency-Accuracy Trade-Off

An analysis of the predictive accuracy versus training time, visualized in Figure 3, suggests a strong connection between model sparsity and overall performance.

The **Stepwise Regression** model provided the most balanced solution, achieving near-best accuracy with relatively low training time, making it the most efficient choice.

While the **Lasso** and **Elastic Net** models yielded the highest accuracy (lowest test RMSE $\approx 3.35 - 3.40$), they incurred the second and third longest training times, respectively. Conversely, **Ridge** and **PCR** demonstrated the fastest training times but were not competitively accurate.

The **PLS** model performed the worst overall, exhibiting both the longest training duration and the lowest predictive performance.

V. Conclusion(s)

This study concludes that the **Lasso** and **Elastic Net** models provide the optimal trade-off between **predictive accuracy** and **computational efficiency**. These methods successfully generated **sparse and effective** models.

While **Stepwise selection** is relatively fast (efficient training), it consistently yielded slightly lower accuracy than the top-performing Lasso and Elastic Net.

The **dimension reduction** techniques, Partial Least Squares (PLS) and Principal Component Regression (PCR), were generally less competitive. Contrary to typical assumptions, **PCR** demonstrated superior performance over **PLS** in both accuracy and training speed. The **Ridge** model offered intermediate performance and was not deemed useful in this context.

In summary, **Lasso or Elastic Net** are the recommended models for robust predictive performance. **Stepwise selection** should only be favored when **training time is the absolute critical constraint**.

APPENDIX

APPENDIX A: SUPPLEMENTARY MATERIALS AND SOFTWARE USAGE

Source Code Repository

The source code, data, and detailed implementation scripts for this project are available on GitHub (URL: https://github.com/gaguillen4384-dev/Data-Science/tree/main/DataMining_1/Project_2).

Software Usage Acknowledgment

The text and documentation were edited and refined using Google Gemini.

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